A New Extension of the Generalized Pareto Distribution for Modeling Low Threshold Exceedances

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Outline

1 **Motivation**
What are the natural hazards? And why do they need proper modeling?

2 **Extreme Value Theory**
Block maxima & Peak-Over-Threshold

3 **Proposed Model**
eGPD via truncated logistic generator

4 **Simulation Study**
Quantile estimation benchmarks

5 **Real Application**
Montréal–Trudeau temperatures

6 **Concluding Remarks**
Summary and future work

Motivation

*Why do we need to go for **natural hazards** modeling?*

What are the natural hazards.

A **natural process** is a sequence of events occurring in nature without human intervention.

Heavy Rainfall – flash floods & catchment overflows

Floods & Hurricanes – large-scale weather events

Avalanches – sudden snow release on slopes

Epidemics – outbreaks such as influenza or COVID-19

The Core Challenge

- Accurate statistical modeling of these **rare events** is vital for understanding and preparing for hazards in advance.
- Standard models describe *average* behavior but often fail for extremes.

Role of Extreme Value Theory (EVT)

- Many real-world decisions depend on **rare, high-impact events**:
  - **Flood risk:** maximum annual river level informs dam design
  - **Finance:** extreme market losses drive risk management
  - **Climate science:** heatwaves, heavy rainfall
- We rarely observe such events in our data. We are generally interested in estimating them **beyond the range of the observation**.
- The quantities are called **return levels** associated with the **return period**.
- EVT provides a framework for modeling **rare events** and for estimating **return levels** in two main approaches:

Block Maxima | **Peak-Over-Threshold (POT)**

Block Maxima

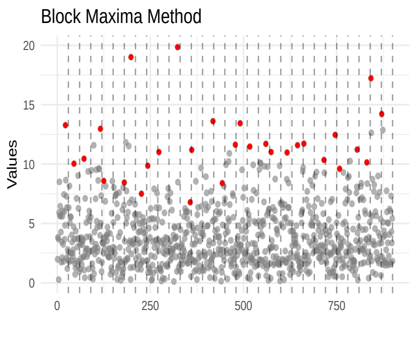
Block Maxima

Divide data into equal-length blocks (e.g. monthly) and take the maximum from each block, i.e., $Y = \max\{X_{i1}, X_{i2}, \ldots, X_{in}\}$.

GEV distribution (Fisher and Tippett, 1928; Gnedenko, 1943):

$$G(y) = \exp\Big\{ -\big[1 + \gamma \tfrac{y-\mu}{\sigma}\big]^{-1/\gamma} \Big\}$$

\+ **Advantage:** simple, theoretically justified.
– **Limitation:** only 1 obs. per block.



Peak-Over-Threshold

Peak-Over-Threshold (POT)

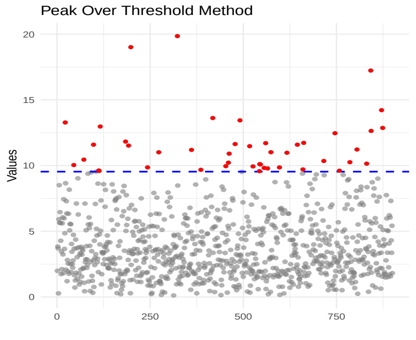
Focus on all observations exceeding a threshold u:

$$Y = X - u \mid X > u$$

GPD distribution (Balkema and Haan, 1974):

$$\mathcal{H}(y) = 1 - \left(1 + \frac{\gamma y}{\sigma}\right)^{-1/\gamma}$$

+ **Advantage:** more extreme data used.
! **Challenge:** choosing u – too low and GPD fails!



The Low Threshold Problem

When does the GPD break down?

Low u
more data available

!
GPD misspecification

High u
too few exceedances

Although theoretically sound, the GPD often fails to capture both **moderate and extreme data simultaneously**, especially when the threshold is set **low enough to ensure sufficient data for inference.**

Our solution: an Extended GPD (eGPD) that models the full distribution above a low threshold – preserving the GPD tail while adding flexibility in the bulk.

Proposed Model: Probability Integral Transform

1. **Integral transform.** $Y = \mathcal{H}^{-1}(W; \boldsymbol{\theta})$, $W \in [0,1]$ with unknown cdf $K_W(\cdot; \boldsymbol{\psi})$ (Papastathopoulos and Tawn, 2013).
2. **CDF of Y.** $\mathcal{F}(y; \boldsymbol{\theta}, \boldsymbol{\psi}) = \mathbb{P}(Y \leq y) = K_W\big(\mathcal{H}(y; \boldsymbol{\theta}); \boldsymbol{\psi}\big)$
3. **PDF of Y.** $f(y; \boldsymbol{\theta}, \boldsymbol{\psi}) = k_W\big(\mathcal{H}(y; \boldsymbol{\theta}); \boldsymbol{\psi}\big)\, h(y; \boldsymbol{\theta})$, with $\boldsymbol{\theta} = (\sigma, \gamma)$.

Tail-preservation conditions on K_W (Gamet and Jalbert, 2022; Naveau et al., 2016):

$C_1 : \lim_{w \uparrow 1} k_W(w) = a > 0 \quad | \quad C_2 : k_W(0) > 0 \quad | \quad C_3 : \lim_{w \uparrow 1} k'_W(w) = 0$

Logistic Distribution and Truncation

Logistic distribution

Let $X \sim \mathrm{Logistic}(1, 1/\kappa)$. Its cdf and pdf are

$$F(x) = \frac{1}{1 + e^{-\kappa(x-1)}},$$

$$f(x) = \frac{\kappa e^{-\kappa(x-1)}}{\left(1 + e^{-\kappa(x-1)}\right)^2}, \qquad x \in \mathbb{R}.$$

Truncation to $[0, 1]$

Restrict the logistic random variable to the interval $[0, 1]$,

$$W = X \mid 0 \le X \le 1.$$

The truncated cdf is

$$K_W(w) = \frac{F(w) - F(0)}{F(1) - F(0)}, \qquad 0 \le w \le 1.$$

The normalizing constant is

$$Z = F(1) - F(0).$$

The Truncated Logistic (TL) Generator

TL distribution on $[0,1]$

CDF:

$$K_W(w;\kappa) = \frac{1}{Z}\left(\frac{1}{1+e^{-\kappa(w-1)}} - \frac{1}{1+e^{\kappa}}\right)$$

PDF:

$$k_W(w;\kappa) = \frac{1}{Z}\,\frac{\kappa e^{-\kappa(w-1)}}{[1+e^{-\kappa(w-1)}]^2}$$

where $Z = 0.5 - \frac{1}{1+e^{\kappa}}$.

Conditions verified:

$C_1: \ \lim_{w\uparrow 1} k_W = \frac{\kappa}{4Z} > 0$
$C_2: \ k_W(0) = \frac{\kappa e^{\kappa}}{Z(1+e^{\kappa})^2} > 0$
$C_3: \ \lim_{w\uparrow 1} k'_W(w) = 0$

κ controls bulk shape

- $\kappa \to 0$: eGPD reduces to GPD
- small κ: flat, uniform-like bulk
- large κ: logistic mode near 1
- smooth bulk-to-tail transition

eGPD: Closed-Form Expressions

CDF:

$$\mathcal{F}(y;\sigma,\gamma,\kappa) = \frac{1}{Z}\left(\frac{1}{1+e^{-\kappa(\mathcal{H}(y)-1)}} - \frac{1}{1+e^{\kappa}}\right)$$

PDF:

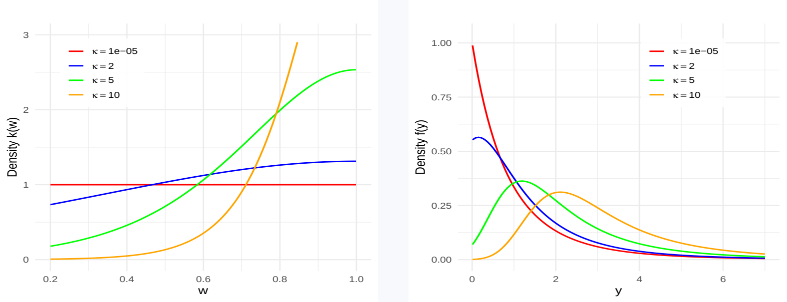
$$f(y;\sigma,\gamma,\kappa) = \frac{1}{Z}\left(\frac{\kappa e^{-\kappa(\mathcal{H}(y)-1)}}{[1+e^{-\kappa(\mathcal{H}(y)-1)}]^2}\right)\cdot\frac{d}{dy}\mathcal{H}(y;\sigma,\gamma)$$

$\kappa > 0 =$ bulk, $\sigma > 0 =$ scale, $\gamma =$ tail

Regularly varying tail check: $1-\mathcal{F}(y) \sim (\kappa/4Z)(\gamma y/\sigma)^{-1/\gamma}$ as $y \to \infty$

Tail index γ is preserved identically to the GPD. As $\kappa \to 0$, the eGPD reduces to the GPD.

K_W and Proposed eGPD Density Behaviour



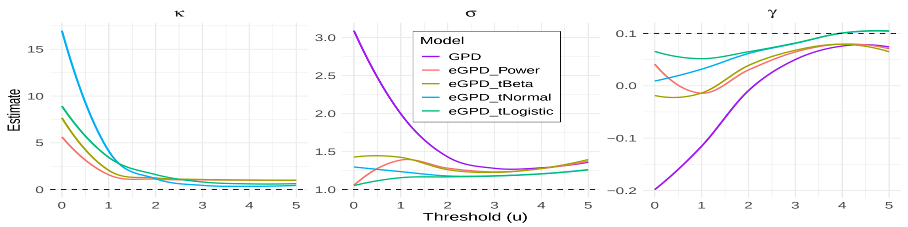
(Left) Density of the truncated logistic distribution on the unit interval as a function of κ, (right) corresponding extension of the generalized Pareto model for fixed $\sigma = 1$, tail index $\gamma = 0.2$.

Simulation Study

Parameter estimates

Setup: $m = 1000$ samples · $n = 3000$ each · GEV$(\mu{=}2, \sigma{=}1, \gamma \in \{0.1, 0.2, 0.3\})$ · Target $p = 1 - 1/10{,}000$

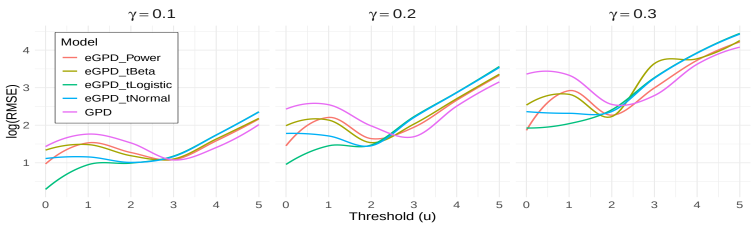
Models compared: **eGPD_tLogistic**, eGPD_tNormal, eGPD_tBeta, eGPD_Power, GPD.



Simulation Study

High-quantile estimation across varying thresholds

Estimated high quantiles comparison:



Result: eGPD_tLogistic has the **lowest RMSE for thresholds** < 2 across all γ, aligns with eGPD_tNormal beyond 2, and all models reduce to the GPD above threshold 3.

Real Application: Montréal–Trudeau Airport

Summer maximum temperatures · June–August 2000–2020

Location: Montréal–Trudeau Int'l Airport, Québec, Canada

Period: June–August, 2000–2020 (summer temprature)

Variable: daily maximum temperature ($^\circ$C)

Thresholds: $u \in \{22, 23, 24, 28\}^\circ$C (exceedance rate 91%–27%)

Model assessment criteria

- BIC
- KS test p-values via Monte Carlo bootstrap
- QQ-plots with 95% bootstrap CIs
- Return-level curves (10- & 100-year events)

Results: BIC and Goodness-of-Fit (KS p-values)

BIC with KS bootstrap p-values in parentheses – lower BIC = better; p-value > 0.05 = acceptable fit

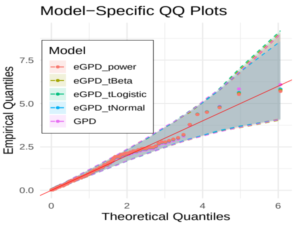
u	eGPD_tLogistic	eGPD_tNormal	eGPD_tBeta	GPD
22°C	**917.24 (0.978)**	917.38 (0.965)	918.00 (0.821)	919.11 (0.013)
23°C	**1065.00 (0.949)**	1065.06 (0.940)	1066.13 (0.739)	1070.05 (0.032)
24°C	1191.89 (0.803)	**1191.82 (0.842)**	1193.21 (0.726)	1200.05 (0.018)
28°C	800.59 (0.852)	800.55 (0.848)	799.85 (0.813)	**795.77 (0.837)**

Green = best at that threshold · **red = GPD; p-value < 0.05 at low thresholds means poor fit**

Low thresholds (22–23°C): eGPD_tLogistic wins on BIC; GPD p-values indicate poor fit.
High threshold (28°C): all models converge ⇒ GPD becomes optimal.

Diagnostic Plots: QQ-Plots & Return Levels

QQ-plot at $u = 28°$C (high)



Return levels (eGPD_tLogistic, $u = 22°$C): $\approx 33°$C once in 10 years · $\approx 36°$C once in 100 years

Concluding Remarks

Model

- Novel eGPD via truncated logistic generator
- Preserves GPD tail index γ exactly
- κ controls bulk flexibility
- Density positive at origin; smooth transition

Performance

- Outperforms eGPD competitors at low thresholds
- Lowest RMSE for $p = 1 - 1/10{,}000$
- Best BIC & KS fit for Montréal (22–23°C)
- Converges to GPD at high thresholds

Future Work

- Spatio-temporal extension using GAMs
- Alternative generator choices for K_W
- Spatial & temporal dependence structures

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Thank you for your attention!

Questions & comments welcome

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